STAT A411/611 Class Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Plan and Review

Today: Sections 3.5–3.8.
Monday: Sections 4.1 and 4.2.
Review: Basics of discrete random variables, Sections 3.1–3.3.

Sections 3.1–3.3: Discrete Random Variables

$$X, \quad \mathcal{X}, \quad \text{pmf}, \quad \text{cdf}, \quad E[X], \quad \text{sd}[X].$$

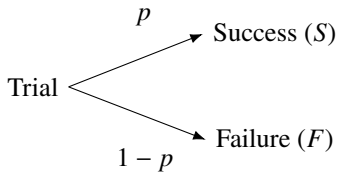
Section 3.4: Discrete Uniform Distribution

If the k possible values $x_1, x_2, \ldots, x_k$ are equally likely, then

$$\boxed{X \sim U(k), \qquad P(X = x_i) = \frac{1}{k}, \quad i = 1, 2, \ldots, k.}$$

Section 3.4: Geometric Distribution

Consider a sequence of identical, successive, independent Bernoulli trials. Each trial has two possible outcomes:



Let X be the number of trials needed until the first success appears. Then $X \sim G(p)$. The formulas are:

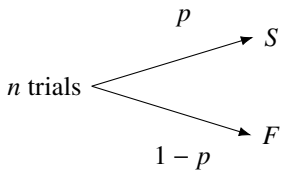
$$\begin{aligned}
\text{pmf:} \quad & P(X = x) = (1-p)^{x-1}p, \quad x = 1, 2, 3, \ldots, \\
\text{cdf:} \quad & F(x) = 1 - (1-p)^{\lfloor x \rfloor}, \quad x \geq 1, \\
& F(x) = 0, \quad x < 1, \\
& E[X] = \frac{1}{p}, \qquad \text{sd}[X] = \frac{\sqrt{1-p}}{p}.
\end{aligned}$$

As $p = P(\text{success})$ increases, fewer trials are expected before the first success. Thus $E[X]$ decreases and is inversely proportional to p.

Chapter 3 - Discrete Random Variables and Their Probability Distributions

Section 3.5 - The Binomial Distribution

Perform n independent Bernoulli trials, each with



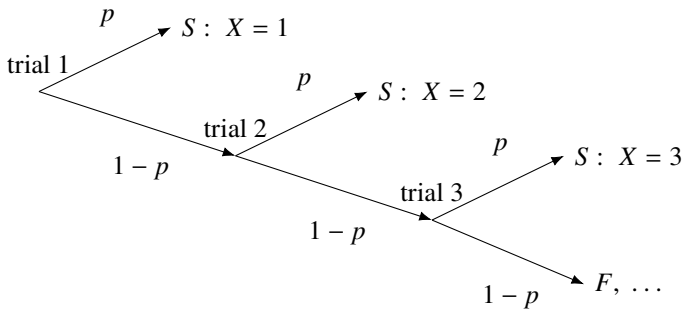
Let X be the **number of successes in the n trials**. Then

$$\boxed{X \sim B(n, p), \qquad \mathcal{X} = \{0, 1, 2, \ldots, n\}.}$$

Geometric and Binomial Distributions Compared

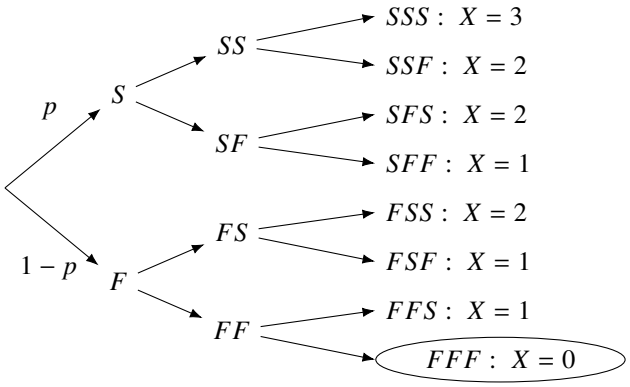
Distribution	Number of trials	Number of successes	$\mathcal{X}$
$G(p)$	X (random)	1	$\{1, 2, 3, \ldots\}$
$B(n, p)$	n (fixed)	X (random)	$\{0, 1, \ldots, n\}$

For $G(p)$, a failure leads to another trial. Success ends the experiment:



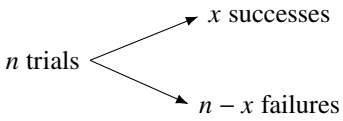
Example: Three Binomial Trials, $X \sim B(3, p)$

For three trials, there are $2^3 = 8$ possible success/failure sequences:



Binomial Probability Mass Function

For $X \sim B(n, p)$, calculate $P(X = x)$. Out of n trials:



One possible sequence has the arrangement

$$\underbrace{S\,S\,\cdots\,S}_{x \text{ successes}}\ \underbrace{F\,F\,\cdots\,F}_{n-x \text{ failures}}.$$

The probability of any specified arrangement with these counts is

$$\underbrace{p\cdots p}_{x \text{ times}}\ \underbrace{(1-p)\cdots(1-p)}_{n-x \text{ times}} = p^x(1-p)^{n-x}.$$

The number of distinct arrangements is

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Therefore the **binomial pmf** is

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{binomial coefficient}} p^x(1-p)^{n-x}, \quad x = 0, 1, \ldots, n.$$

Example: Fifteen Rolls of a Fair Die

Roll a fair six-sided die 15 times. Find the probability that a **four** appears exactly three times.



Let X be the number of fours in 15 rolls. Then

$$X \sim B\left(15, \frac{1}{6}\right), \qquad p = \frac{1}{6}, \qquad 1 - p = \frac{5}{6}.$$

For any specified order with three fours and twelve non-fours, the probability is

$$\left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{12}.$$

There are $\binom{15}{3}$ possible orders. Thus

$$\begin{aligned} P(X = 3) &= \binom{15}{3}\left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{12} \\ &= \frac{15!}{3!12!} \cdot \frac{5^{12}}{6^{15}} \\ &= \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1} \cdot \frac{5^{12}}{6^{15}} \\ &= \boxed{455 \cdot \frac{5^{12}}{6^{15}}}. \end{aligned}$$

Mean and Standard Deviation of a Binomial Random Variable

If $X \sim B(n, p)$, then

$$\boxed{E[X] = np}, \qquad \boxed{\text{sd}[X] = \sqrt{np(1-p)}}.$$

Does $E[X]$ being directly proportional to n and p make sense?

- If n increases and p stays the same, more successes are expected; $E[X]$ increases.
- If p increases and n stays the same, more successes are expected; $E[X]$ increases.

Using the Binomial CDF Table

Table I (p. 690) gives the cdf for $X \sim B(n, p)$ with

$$n = 5, 6, 7, \ldots, 20, \qquad p = 0.1, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 0.9.$$

For $X \sim B(17, 0.3)$, find:

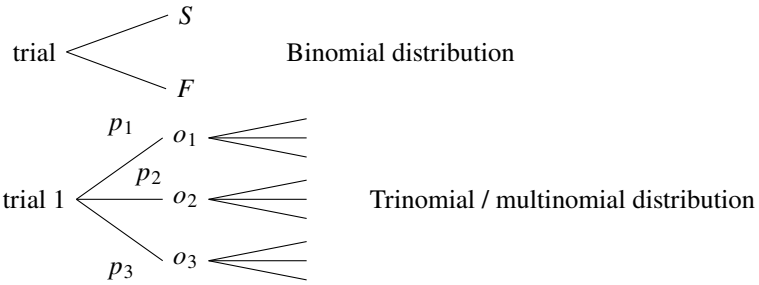
a) $P(X \leq 4) = \boxed{0.3887}$.

b) $P(X = 5) = P(X \leq 5) - P(X \leq 4) = 0.5968 - 0.3887 = \boxed{0.2081}$.

c) $P(1 \le X \le 8) = P(X \le 8) - P(X \le 0) = 0.9597 - 0.0023 = \boxed{0.9574}$.

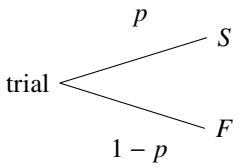
Binomial, Trinomial, and Multinomial Distributions

A Bernoulli trial has two possible outcomes, S and F. More generally, a trial can have three possible outcomes, o_1, o_2, o_3, with probabilities p_1, p_2, p_3.



Section 3.6 - The Negative Binomial Distribution

Perform Bernoulli trials until reaching the rth success ($r = 1, 2, 3, \ldots$):



Let X be the number of trials needed to reach the rth success. The notation is

$$\boxed{X \sim NB(r, p)}, \qquad \boxed{NB(1, p) = G(p)}.$$

The possible values of X are

$$\mathcal{X} = \{r, r+1, r+2, \ldots\}.$$

For $X = x$, the rth success must occur on trial x. The first $x - 1$ trials have $r - 1$ successes and $x - r$ failures; the last trial is a success:

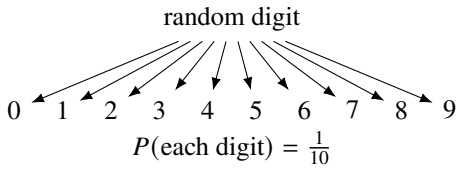
$$\underbrace{\binom{x-1}{r-1} p^{r-1}(1-p)^{x-r}}_{\text{first } x-1 \text{ trials}} \underbrace{p}_{\text{last trial}}.$$

Hence the **negative-binomial pmf** is

$$\boxed{P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, \ldots.}$$

Example: Random Digits

A random-number generator produces digits $0, 1, \ldots, 9$, each with probability $1/10$.



For example, a sequence of digits might be

$$4\,4\,7\,\underline{0}\,8\,8\,8\,5\,2\,1\,2\,5\,6\,9\,\underline{0}\,\ldots\,\underbrace{0}_{\text{40th digit}}.$$

Find the probability that the **fifth zero** appears in the 40th generated digit. Let X be the number of digits generated until the fifth zero; then $X \sim NB(5, 1/10)$. For $X = 40$, the first 39 digits must contain four zeros and 35 nonzeros, followed by a zero:

$$\begin{aligned} P(X = 40) &= \binom{39}{4}\left(\frac{1}{10}\right)^4\left(\frac{9}{10}\right)^{35}\left(\frac{1}{10}\right) \\ &= \boxed{\binom{39}{4}\left(\frac{1}{10}\right)^5\left(\frac{9}{10}\right)^{35}}. \end{aligned}$$

Mean of a Negative Binomial Random Variable

If $X \sim NB(r, p)$, then

$$\boxed{E[X] = \frac{r}{p}}.$$

How many digits are expected before reaching the seventh zero? Let X be the total digits generated until the seventh zero. Then

$$X \sim NB\left(7, \frac{1}{10}\right), \qquad E[X] = \frac{7}{1/10} = \boxed{70}.$$

$$\underbrace{d_1, d_2, \ldots, d_{69}}_{\substack{\text{69 digits} \\ \text{6 zeros and 63 others}}} \qquad \underbrace{0}_{\text{70th digit: seventh zero}}.$$

Section 3.7 - The Hypergeometric Distribution

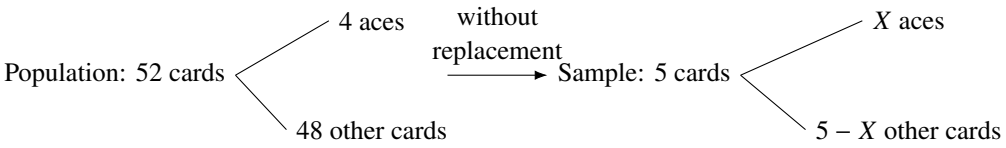
Start with a population of N elements, of which M have a particular characteristic ("successes"). Take a random sample of n elements **without replacement**. Let X be the number of successes in the sample.

$$\boxed{X \sim HG(N, M, n)}.$$

Example: Poker Hand

A deck of 52 cards has four aces and 48 other cards. The first player receives five cards from the dealer. Let X be the number of aces in that hand:

$$\boxed{X \sim HG(52, 4, 5)}.$$



If $X \sim HG(N, M, n)$, the **hypergeometric pmf** is

$$\boxed{P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}}.$$

Find the probability that the first player has **exactly one ace**:

$$\boxed{P(X = 1) = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}}}.$$

The number of ways to select four non-aces is

$$\binom{48}{4} = \frac{48!}{4!\,44!} = \frac{48 \cdot 47 \cdot 46 \cdot 45}{4 \cdot 3 \cdot 2 \cdot 1}.$$

Mean of a Hypergeometric Random Variable

For $X \sim HG(N, M, n)$,

$$\boxed{E[X] = n\frac{M}{N}} \qquad \text{(expected number of successes)}.$$

Section 3.8 - The Poisson Distribution

Definition. A random variable X has the Poisson distribution with parameter λ when

$$\boxed{X \sim P(\lambda)}, \qquad \boxed{P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}}, \quad x = 0, 1, 2, \ldots.$$

Here $e \approx 2.71828\ldots$ is Euler's number. The main properties are

$$\boxed{E[X] = \lambda}, \qquad \boxed{V[X] = \lambda}.$$

Poisson probabilities appear in Table II.

Poisson Process

Events such as earthquakes, accidents, and emergency-room arrivals may follow a Poisson process with rate α events per unit time. Let X count the number of events in an interval of length t time units. Then

$$\boxed{X \sim P(\alpha t)}.$$

Examples of counts modeled by a Poisson process:

- Number of earthquakes in one year in the Pacific Northwest.
- Number of accidents on the Glenn Highway in one month.
- Number of patients arriving at an emergency room in a fixed time interval.

Suppose the mean number of ER arrivals between 5 PM and 6 PM, over many days, is 6.9. Let

$$X = \text{number of ER patients arriving tomorrow between 5 PM and 6 PM.}$$

Then

$$\boxed{X \sim P(6.9)}.$$

Distribution Summary

Section	Distribution	Notation
3.4	Discrete uniform / Geometric	$U(k),\ G(p)$
3.5	Binomial	$B(n, p)$
3.6	Negative binomial	$NB(r, p)$
3.7	Hypergeometric	$HG(N, M, n)$
3.8	Poisson	$P(\lambda)$

Section 3.9 is skipped. The next class begins Chapter 4 (Sections 4.1–4.2), with continuous distributions including the normal, exponential, gamma, and chi-squared families.